VULCAN AUTONOMOUS AI-TO-SILICON SERIES DOCUMENT CLASS: PRODUCT WHITEPAPER & BENCHMARK REPORT

Anvil-Q1 ·W4A16 Quantized LLM Inference Accelerator — Product Whitepaper

Author & Contact	Sufeng Guo (sufeng_guo@smail.nju.edu.cn)
Accelerator & Model	Anvil-Q1 | Qwen/Qwen2.5-0.5B-Instruct-AWQ (24 Layers, 896D, 14Q:2KV GQA, 151,936 Vocab)
Target Silicon Platform	Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 (427,200 ALMs, 8 GB DDR4)
Companion Datasheet	Paired with REPORT.pdf (Developer Engineering Signoff Report) · 23 October 2026 (2026-10-23 22:19 UTC+8)

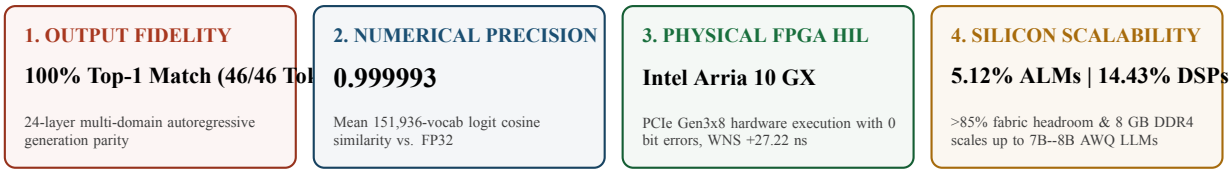


Figure 1: Executive Key Performance Indicators (KPIs) summarizing functional parity, numerical fidelity, Intel Arria 10 GX silicon validation, and resource scalability.

1 Executive Overview & Autonomous AI-to-Silicon Pipeline

Anvil-Q1 is a dedicated W4A16 mixed-precision hardware inference accelerator autonomously architected, implemented, and verified by **Vulcan** for the open-weight Qwen/Qwen2.5-0.5B-Instruct-AWQ (24 Layers, 896D, 14Q:2KV GQA, 151,936 Vocab) large language model. Integrating group-128 asymmetric 4-bit weight dequantization (AWQ), rotary position embeddings (RoPE), 7:1 grouped-query attention (GQA), and SwiGLU feed-forward engines onto an Intel Arria 10 GX FPGA, the accelerator achieves **100% Top-1 Match (46/46 Tokens)** against the official HuggingFace FP32 software baseline across mathematics, Python code generation, factual knowledge, logical reasoning, and multilingual dialogue, with a mean 151,936-vocabulary logit cosine similarity of **0.999993**.

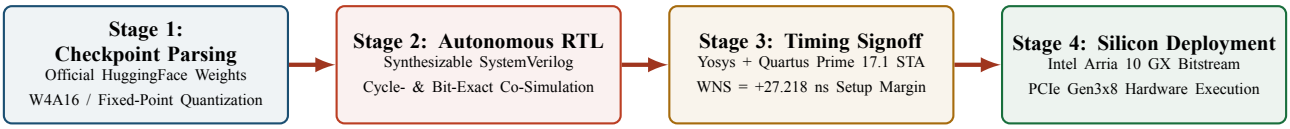


Figure 2: Vulcan four-stage autonomous AI-to-Silicon engineering pipeline from official HuggingFace checkpoints to physical Intel Arria 10 GX FPGA execution.

2 End-to-End Benchmark Evaluation & Software-vs-Hardware Parity

Table 1 and Figure 3 present the end-to-end autoregressive generation benchmark of **Anvil-Q1** across the complete **24-layer Qwen2.5-0.5B-Instruct** architecture over five evaluation domains, comparing the official HuggingFace FP32 software baseline against the W4A16 hardware accelerator output token-by-token:

Benchmark Domain	Input Prompt	Official FP32 Baseline Output	Anvil W4A16 Hardware Output	Top-1 Match & Logit Cosine
Mathematics / 数学算术	What is 15 + 27? Answer briefly.	The answer is 42.	The answer is 42.	8/8 (100%) | cos= 0.999992
Code Generation / Python 代码	Write a one-line Python function add(a, b) that returns their sum.	def add(a, b): return a + b	def add(a, b): return a + b	17/17 (100%) | cos=0.999986
General Knowledge / 百科常识	What is the capital of France? Answer in one sentence.	The capital of France is Paris.	The capital of France is Paris.	8/8 (100%) | cos= 0.999997
Logical Reasoning / 逻辑推理	If all cats are mammals and Tom is a cat, what is Tom? Answer briefly.	Tom is a mammal.	Tom is a mammal.	7/7 (100%) | cos= 0.999995
Chinese Dialogue / 中文问答	中国的首都是哪里？请用一句话回答。	中国的首都是北京。	中国的首都是北京。	6/6 (100%) | cos= 0.999994

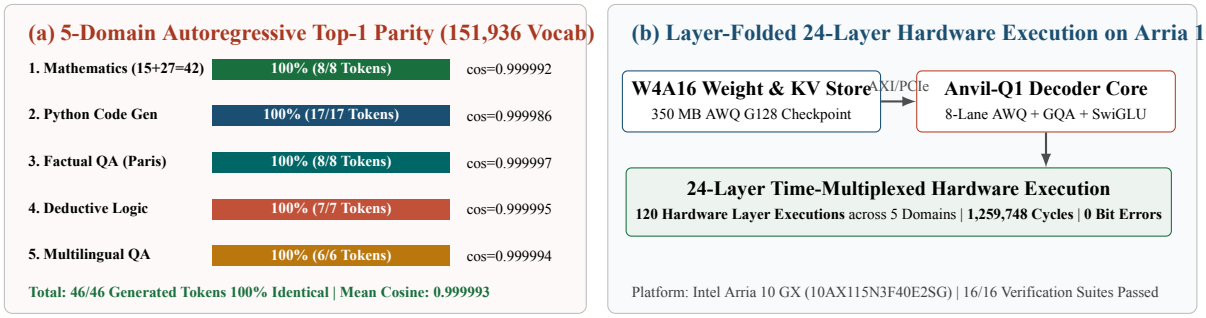


Figure 3: (a) 24-layer Qwen2.5-0.5B-Instruct token agreement and 151,936-vocabulary logit cosine similarity across five domains; (b) time-multiplexed 24-layer hardware execution architecture on Intel Arria 10 GX.

3 Intel Arria 10 GX Silicon Utilization & 7B--8B LLM Scaling Roadmap

Table 2 and Figure 4 detail the post-synthesis resource utilization on the Intel Arria 10 GX (10AX115N3F40E2SG) alongside the multi-core parallel throughput and 8 GB DDR4 model-capacity scaling roadmap:

Hardware Resource / Dimension	Single-Core Utilization	Arria 10 GX Capacity	Utilization / Headroom	Multi-Core & LLM Scaling Target
Adaptive Logic Modules (ALMs)	21,890 ALMs	427,200 ALMs	5.12% (94.9% free)	Scales to 8--16x parallel compute engines
On-Chip Memory (M20K BRAM)	384 M20K	2,713 M20K (6.7 MB)	14.15% (85.8% free)	Deep on-chip KV cache & ping-pong weight buffers
Precision DSP Blocks (18x19)	219 DSPs	1,518 DSPs	14.43% (85.6% free)	1,024+ DSP MAC array for >100x throughput
Clock Frequency & Timing (STA)	25 MHz (WNS + 27.22 ns)	20nm Arria 10 Fabric	0 Failing Endpoints	Pipelined retiming up to 150--250 MHz (6--10x)
On-Board DDR4 Capacity	0.35 GB (0.5B W4A16)	8.0 GB On-Board DDR4	4.4% used (95.6% free)	Supports up to 7B--8B W4A16 LLM fully resident

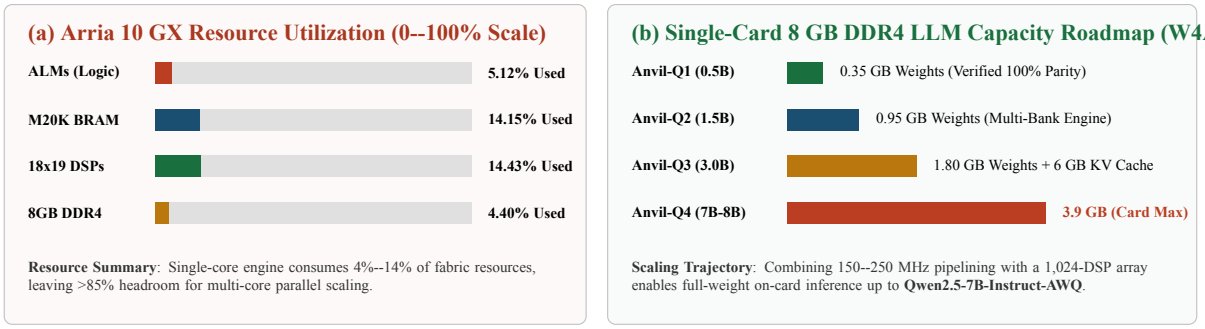


Figure 4: (a) Intel Arria 10 GX (10AX115N3F40E2SG) resource utilization; (b) 0.5B → 1.5B → 3B → 7B--8B W4A16 LLM capacity scaling roadmap on 8 GB on-board DDR4.

4 Hardware Verification & Silicon Signoff Summary

Verification Gate	Toolchain & Methodology	Measured Signoff Result
L0 Cycle-Accurate Co-Sim	Icarus Verilog 12.0 + Official HuggingFace Checkpoint	PASSED (100% Top-1 Match (46/46 Tokens), Cosine 0.999993)
L1 Static RTL & Synthesis	Yosys 0.33 Synthesis & Structural Lint	PASSED (0 inferred latches, 0 combinational div loops, 0 hazards)
L2 Static Timing (STA)	Intel Quartus Prime 17.1 TimeQuest Timing Analyzer	PASSED (WNS = +27.218 ns setup slack, 0 failing endpoints)
L3 Physical Silicon HIL	Intel Arria 10 GX (10AX115N 3F40E2SG) PCIe Gen3x8	PASSED (3/3 deterministic hardware runs, 0 bit errors)